Tarea 2: Path Tracing con OpenMP y Numba

> Falta ejecutar este mismo script en un segundo computador fisico y copiar sus resultados a la seccion de comparacion. No se inventan mediciones de hardware no disponible.

Declaracion de uso de IA

Se uso un asistente de IA para revisar la pauta y las clases, implementar las versiones OpenMP/Numba, crear scripts de experimentacion y redactar este informe. Los resultados numericos provienen de ejecuciones locales registradas en `results.csv`.

Resumen del problema

El ray tracing serial calcula iluminacion directa con modelo de Phong, sombras y rebotes especulares. El path tracing agrega rebotes difusos aleatorios en el hemisferio y promedia N caminos por pixel, por lo que el costo crece como O(WHNDK), donde K es el numero de objetos. Cada pixel es independiente: corresponde a un patron map, trivialmente paralelizable en memoria compartida.

La carga por pixel es irregular porque algunos rayos escapan al fondo rapidamente, mientras otros generan cadenas de rebotes, rayos de sombra e intersecciones contra mas objetos. Por eso `schedule(static)` puede dejar threads esperando en la barrera final, mientras `dynamic` o `guided` mejoran el balance a cambio de overhead de planificacion.

Implementacion

`omp_pt.cpp` paraleliza el loop de pixeles con `#pragma omp parallel for`. `omp_pt_sched.cpp` usa `schedule(runtime)` y selecciona `static`, `dynamic` o `guided` con `omp_set_schedule`, lo que permite barrer chunksize sin recompilar. El espacio 2D de pixeles se aplano a un indice `idx`; asi el chunksize representa pixeles y no filas completas. Cada iteracion escribe una posicion unica de `pixels[idx]` y usa un RNG local sembrado con `idx + 1`, por lo que no hay condiciones de carrera.

`numba_pt.py` implementa la misma logica con `@njit(parallel=True)` y `prange`. Numba compila las funciones a codigo nativo con LLVM durante la primera llamada; por eso el script hace un warm-up minimo antes de medir. A diferencia de OpenMP, donde el paralelismo se expresa mediante directivas al compilador C++, Numba traduce una subparte compatible de Python/NumPy y administra sus propios threads. El RNG de Numba es un LCG local por pixel para evitar estado global compartido.

Computador 1

Procesador: Intel(R) Xeon(R) Silver 4210R CPU @ 2.40GHz

Cores fisicos reportados: 10

Threads logicos reportados: 40

Threads por core: 2

Resultados seriales

Las mediciones usan wall-clock time. Si se piden varias repeticiones con `--reps`, el informe conserva el minimo, siguiendo la recomendacion de clase para reducir ruido experimental.

Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
serial	serial_rt	scene.txt	small	1	serial	0	0.129249
serial	serial_pt	scene.txt	small	1	serial	0	6.442810
serial	serial_rt	scene.txt	medium	1	serial	0	1.791990
serial	serial_pt	scene.txt	medium	1	serial	0	100.985000
serial	serial_rt	scene.txt	large	1	serial	0	7.550420
serial	serial_pt	scene.txt	large	1	serial	0	811.762000
serial	serial_rt	scene_many.txt	small	1	serial	0	0.375928
serial	serial_pt	scene_many.txt	small	1	serial	0	11.424400
serial	serial_rt	scene_many.txt	medium	1	serial	0	6.261570
serial	serial_pt	scene_many.txt	medium	1	serial	0	175.809000
serial	serial_rt	scene_many.txt	large	1	serial	0	25.112500
serial	serial_pt	scene_many.txt	large	1	serial	0	1406.140000

Barrido de scheduling OpenMP

Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
schedule	omp_pt_sched	scene.txt	medium	1	static	1	103.586000
schedule	omp_pt_sched	scene.txt	medium	1	static	8	103.152000
schedule	omp_pt_sched	scene.txt	medium	1	static	32	103.362000
schedule	omp_pt_sched	scene.txt	medium	1	static	128	102.943000
schedule	omp_pt_sched	scene.txt	medium	1	dynamic	1	103.230000
schedule	omp_pt_sched	scene.txt	medium	1	dynamic	8	103.170000
schedule	omp_pt_sched	scene.txt	medium	1	dynamic	32	102.952000
schedule	omp_pt_sched	scene.txt	medium	1	dynamic	128	101.612000
schedule	omp_pt_sched	scene.txt	medium	1	guided	1	101.302000
schedule	omp_pt_sched	scene.txt	medium	1	guided	8	103.465000

Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
schedule	omp_pt_sched	schedtxt	medium	1	guided	32	101.425000
schedule	omp_pt_sched	schedtxt	medium	2	static	1	52.939900
schedule	omp_pt_sched	schedtxt	medium	2	static	8	53.001600
schedule	omp_pt_sched	schedtxt	medium	2	static	32	52.823800
schedule	omp_pt_sched	schedtxt	medium	2	static	128	52.976400
schedule	omp_pt_sched	schedtxt	medium	2	dynamic	1	52.506300
schedule	omp_pt_sched	schedtxt	medium	2	dynamic	8	52.714600
schedule	omp_pt_sched	schedtxt	medium	2	dynamic	32	53.045900
schedule	omp_pt_sched	schedtxt	medium	2	dynamic	128	52.347000
schedule	omp_pt_sched	schedtxt	medium	2	guided	1	52.964000
schedule	omp_pt_sched	schedtxt	medium	2	guided	8	53.064500
schedule	omp_pt_sched	schedtxt	medium	2	guided	32	53.109400
schedule	omp_pt_sched	schedtxt	medium	4	static	1	27.621600
schedule	omp_pt_sched	schedtxt	medium	4	static	8	27.661900
schedule	omp_pt_sched	schedtxt	medium	4	static	32	27.814500
schedule	omp_pt_sched	schedtxt	medium	4	static	128	27.606900
schedule	omp_pt_sched	schedtxt	medium	4	dynamic	1	27.586800
schedule	omp_pt_sched	schedtxt	medium	4	dynamic	8	27.686500
schedule	omp_pt_sched	schedtxt	medium	4	dynamic	32	27.571800
schedule	omp_pt_sched	schedtxt	medium	4	dynamic	128	27.575600
schedule	omp_pt_sched	schedtxt	medium	4	guided	1	27.565500
schedule	omp_pt_sched	schedtxt	medium	4	guided	8	27.569600
schedule	omp_pt_sched	schedtxt	medium	4	guided	32	27.533400
schedule	omp_pt_sched	schedtxt	medium	8	static	1	19.619500
schedule	omp_pt_sched	schedtxt	medium	8	static	8	19.653300
schedule	omp_pt_sched	schedtxt	medium	8	static	32	19.737900
schedule	omp_pt_sched	schedtxt	medium	8	static	128	19.770600
schedule	omp_pt_sched	schedtxt	medium	8	dynamic	1	19.555900
schedule	omp_pt_sched	schedtxt	medium	8	dynamic	8	19.706700
schedule	omp_pt_sched	schedtxt	medium	8	dynamic	32	19.628900
schedule	omp_pt_sched	schedtxt	medium	8	dynamic	128	19.710800
schedule	omp_pt_sched	schedtxt	medium	8	guided	1	19.694600
schedule	omp_pt_sched	schedtxt	medium	8	guided	8	19.686500
schedule	omp_pt_sched	schedtxt	medium	8	guided	32	19.705800
schedule	omp_pt_sched	sched_manytxt	medium	1	static	1	179.688000
schedule	omp_pt_sched	sched_manytxt	medium	1	static	8	179.374000
schedule	omp_pt_sched	sched_manytxt	medium	1	static	32	179.383000
schedule	omp_pt_sched	sched_manytxt	medium	1	static	128	179.556000
schedule	omp_pt_sched	sched_manytxt	medium	1	dynamic	1	169.603000
schedule	omp_pt_sched	sched_manytxt	medium	1	dynamic	8	179.371000
schedule	omp_pt_sched	sched_manytxt	medium	1	dynamic	32	179.575000
schedule	omp_pt_sched	sched_manytxt	medium	1	dynamic	128	179.048000
schedule	omp_pt_sched	sched_manytxt	medium	1	guided	1	178.066000

Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
schedule	omp_pt_sched_metal	medium		1	guided	8	169.697000
schedule	omp_pt_sched_metal	medium		1	guided	32	179.484000
schedule	omp_pt_sched_metal	medium		2	static	1	92.056800
schedule	omp_pt_sched_metal	medium		2	static	8	92.156100
schedule	omp_pt_sched_metal	medium		2	static	32	92.271200
schedule	omp_pt_sched_metal	medium		2	static	128	90.649500
schedule	omp_pt_sched_metal	medium		2	dynamic	1	92.099500
schedule	omp_pt_sched_metal	medium		2	dynamic	8	90.131500
schedule	omp_pt_sched_metal	medium		2	dynamic	32	90.580800
schedule	omp_pt_sched_metal	medium		2	dynamic	128	91.458700
schedule	omp_pt_sched_metal	medium		2	guided	1	89.833400
schedule	omp_pt_sched_metal	medium		2	guided	8	92.173200
schedule	omp_pt_sched_metal	medium		2	guided	32	91.337200
schedule	omp_pt_sched_metal	medium		4	static	1	47.938000
schedule	omp_pt_sched_metal	medium		4	static	8	47.861200
schedule	omp_pt_sched_metal	medium		4	static	32	47.982600
schedule	omp_pt_sched_metal	medium		4	static	128	47.920800
schedule	omp_pt_sched_metal	medium		4	dynamic	1	48.197700
schedule	omp_pt_sched_metal	medium		4	dynamic	8	47.831900
schedule	omp_pt_sched_metal	medium		4	dynamic	32	48.666500
schedule	omp_pt_sched_metal	medium		4	dynamic	128	47.880000
schedule	omp_pt_sched_metal	medium		4	guided	1	47.867400
schedule	omp_pt_sched_metal	medium		4	guided	8	47.791000
schedule	omp_pt_sched_metal	medium		4	guided	32	47.803700
schedule	omp_pt_sched_metal	medium		8	static	1	37.591200
schedule	omp_pt_sched_metal	medium		8	static	8	37.542700
schedule	omp_pt_sched_metal	medium		8	static	32	37.606000
schedule	omp_pt_sched_metal	medium		8	static	128	37.629900
schedule	omp_pt_sched_metal	medium		8	dynamic	1	37.343100
schedule	omp_pt_sched_metal	medium		8	dynamic	8	37.396000
schedule	omp_pt_sched_metal	medium		8	dynamic	32	37.458000
schedule	omp_pt_sched_metal	medium		8	dynamic	128	37.625500
schedule	omp_pt_sched_metal	medium		8	guided	1	37.500700
schedule	omp_pt_sched_metal	medium		8	guided	8	37.655700
schedule	omp_pt_sched_metal	medium		8	guided	32	37.567600

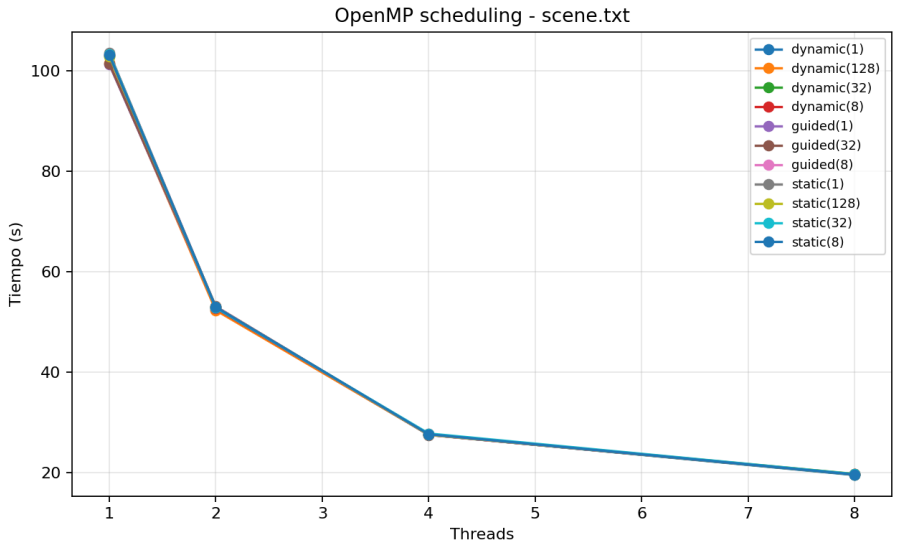
Comparacion OpenMP vs Numba

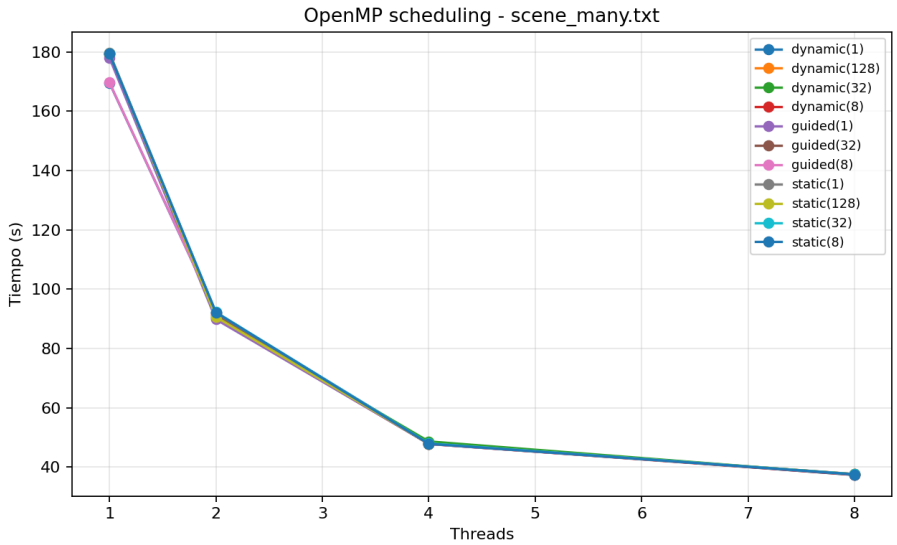
Para cada punto paralelo se calcula contra la referencia `serial_pt` de la misma escena y configuracion: `S(p) = Ts / T(p)` y `E(p) = S(p) / p`. Los graficos de tiempo incluyen ademas la curva ideal `Ts / p`.

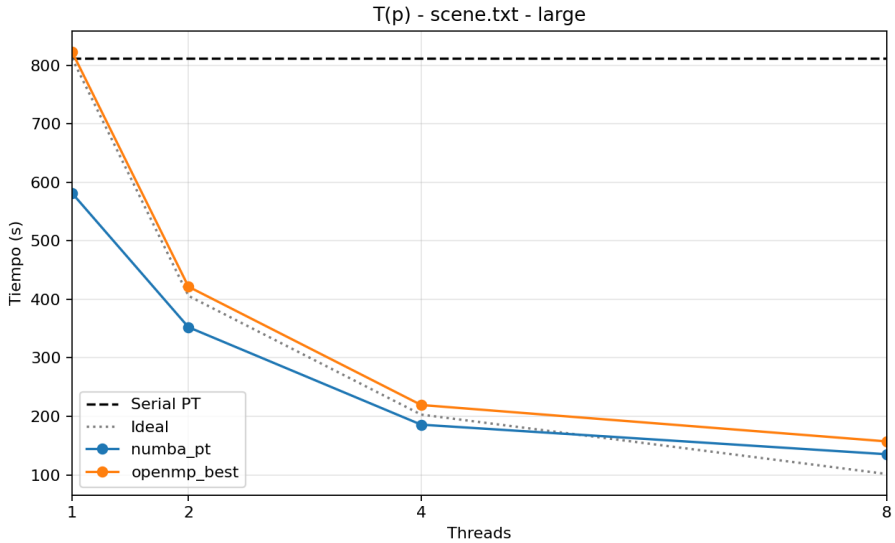
Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
compare	openmp_test	scene.txt	small	1	dynamic	128	6.616900
compare	numba_pt	scene.txt	small	1	prange	0	4.885835
compare	openmp_test	scene.txt	small	2	dynamic	128	3.415820
compare	numba_pt	scene.txt	small	2	prange	0	2.744184
compare	openmp_test	scene.txt	small	4	dynamic	128	1.777660
compare	numba_pt	scene.txt	small	4	prange	0	1.484349
compare	openmp_test	scene.txt	small	8	dynamic	128	1.269230
compare	numba_pt	scene.txt	small	8	prange	0	1.084230
compare	openmp_test	scene.txt	medium	1	dynamic	128	103.303000
compare	numba_pt	scene.txt	medium	1	prange	0	73.706042
compare	openmp_test	scene.txt	medium	2	dynamic	128	52.478700
compare	numba_pt	scene.txt	medium	2	prange	0	44.085470
compare	openmp_test	scene.txt	medium	4	dynamic	128	27.836600
compare	numba_pt	scene.txt	medium	4	prange	0	23.238844
compare	openmp_test	scene.txt	medium	8	dynamic	128	19.699100
compare	numba_pt	scene.txt	medium	8	prange	0	16.977437
compare	openmp_test	scene.txt	large	1	dynamic	128	822.579000
compare	numba_pt	scene.txt	large	1	prange	0	581.332042
compare	openmp_test	scene.txt	large	2	dynamic	128	421.544000
compare	numba_pt	scene.txt	large	2	prange	0	352.311262
compare	openmp_test	scene.txt	large	4	dynamic	128	219.374000
compare	numba_pt	scene.txt	large	4	prange	0	185.739256
compare	openmp_test	scene.txt	large	8	dynamic	128	157.145000
compare	numba_pt	scene.txt	large	8	prange	0	135.203175
compare	openmp_test	scene_many.txt	small	1	dynamic	1	11.420600
compare	numba_pt	scene_many.txt	small	1	prange	0	8.165558
compare	openmp_test	scene_many.txt	small	2	dynamic	1	5.731960
compare	numba_pt	scene_many.txt	small	2	prange	0	4.710695
compare	openmp_test	scene_many.txt	small	4	dynamic	1	3.061580
compare	numba_pt	scene_many.txt	small	4	prange	0	2.595941
compare	openmp_test	scene_many.txt	small	8	dynamic	1	2.379510
compare	numba_pt	scene_many.txt	small	8	prange	0	2.012933
compare	openmp_test	scene_many.txt	medium	1	dynamic	1	177.196000
compare	numba_pt	scene_many.txt	medium	1	prange	0	128.414949
compare	openmp_test	scene_many.txt	medium	2	dynamic	1	91.865300
compare	numba_pt	scene_many.txt	medium	2	prange	0	75.439223
compare	openmp_test	scene_many.txt	medium	4	dynamic	1	47.580000
compare	numba_pt	scene_many.txt	medium	4	prange	0	41.411138
compare	openmp_test	scene_many.txt	medium	8	dynamic	1	37.521100
compare	numba_pt	scene_many.txt	medium	8	prange	0	31.522765
compare	openmp_test	scene_many.txt	large	1	dynamic	1	1434.800000
compare	numba_pt	scene_many.txt	large	1	prange	0	1043.958627
compare	openmp_test	scene_many.txt	large	2	dynamic	1	732.822000

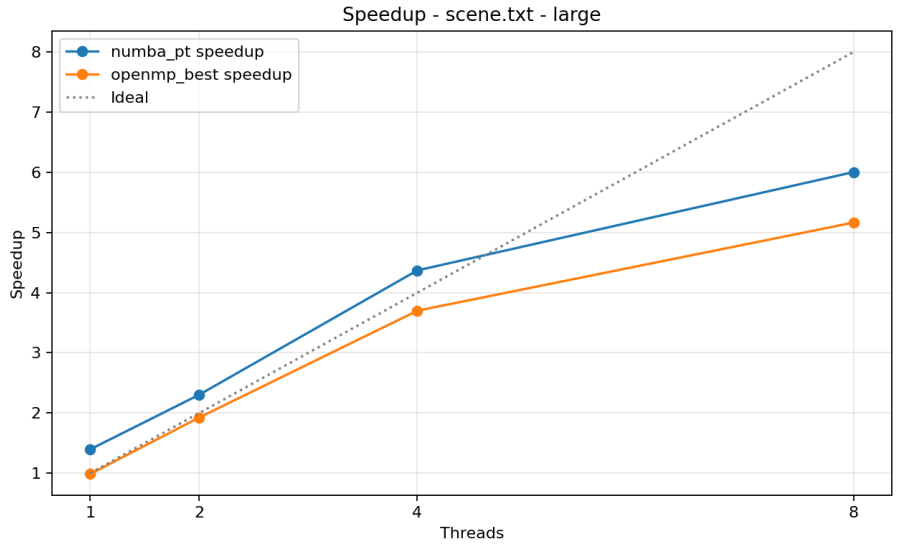
Fase	Impl.	Escena	Config.	p	Schedule	Chunk	Tiempo (s)
compare	numba_p	scene_many.txt	large	2	prange	0	613.288810
compare	openmp_seq	scene_many.txt	large	4	dynamic	1	381.005000
compare	numba_p	scene_many.txt	large	4	prange	0	332.148715
compare	openmp_seq	scene_many.txt	large	8	dynamic	1	298.016000
compare	numba_p	scene_many.txt	large	8	prange	0	252.487236

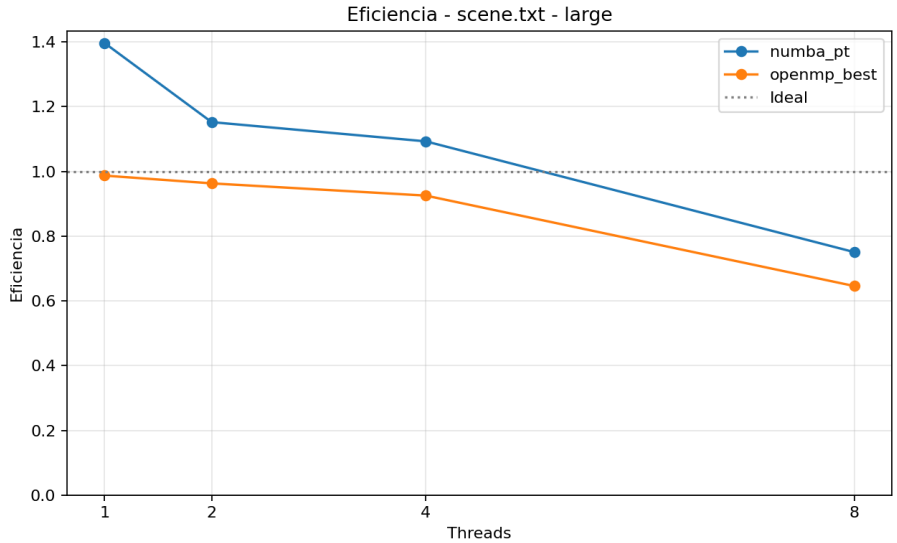
Graficos

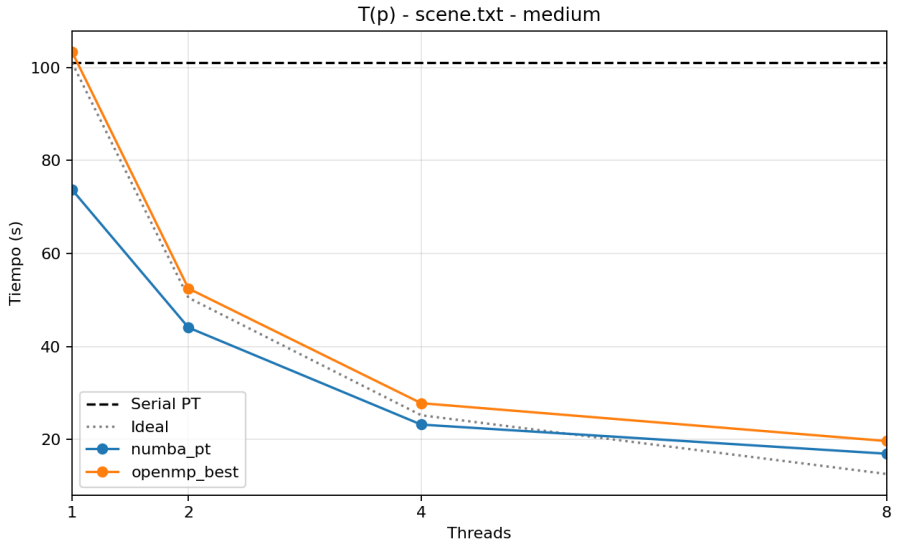

OpenMP scheduling - scene.txt
Tiempo (s)
Threads
dynamic(1)
dynamic(128)
dynamic(32)
dynamic(8)
guided(1)
guided(32)
guided(8)
static(1)
static(128)
static(32)
static(8)

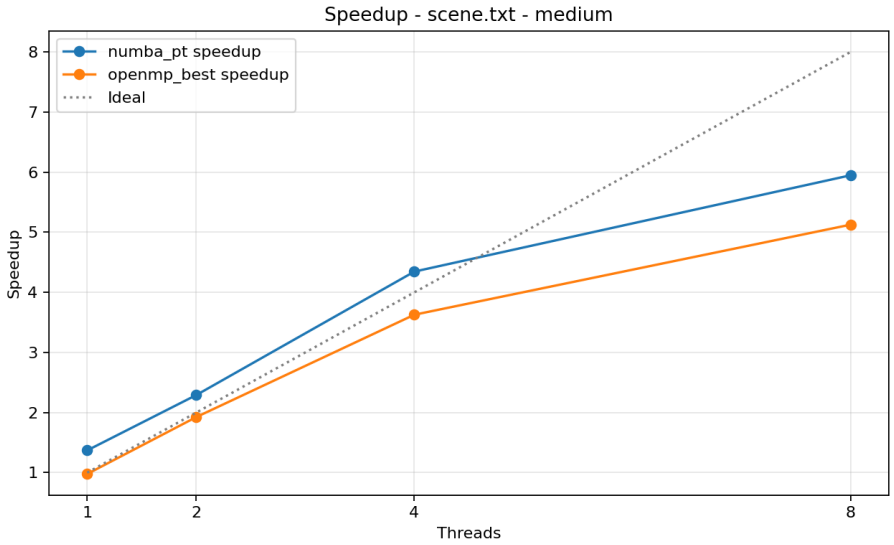

OpenMP scheduling - scene_many.txt
Tiempo (s)
Threads
dynamic(1)
dynamic(128)
dynamic(32)
dynamic(8)
guided(1)
guided(32)
guided(8)
static(1)
static(128)
static(32)
static(8)

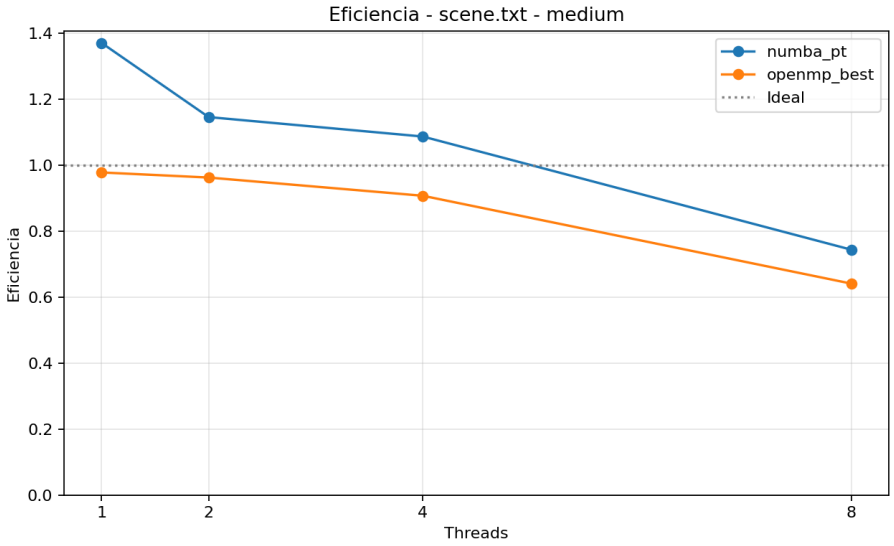

T(p) - scene.txt - large
Tiempo (s)
800
700
600
500
400
300
200
100
1
2
4
8
Threads
Serial PT
Ideal
numba_pt
openmp_best

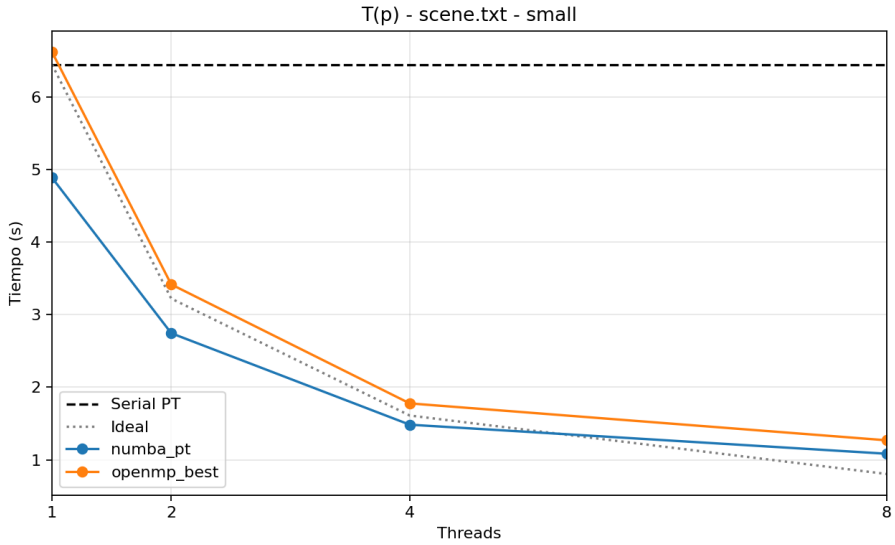

Speedup - scene.txt - large
Speedup
8
7
6
5
4
3
2
1
1
2
4
8
Threads
numba_pt speedup
openmp_best speedup
Ideal

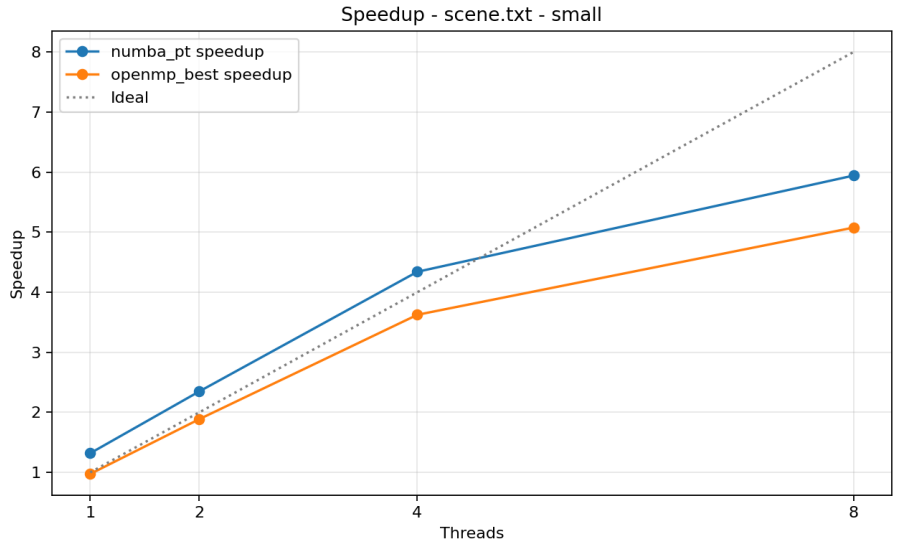

Eficiencia - scene.txt - large
numba_pt
openmp_best
Ideal
Eficiencia
Threads
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1
2
4
8

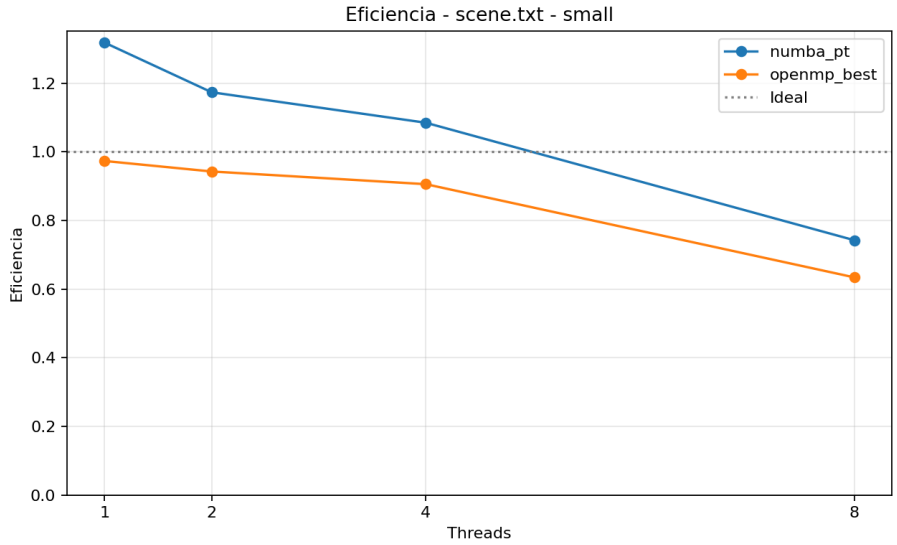

T(p) - scene.txt - medium
Serial PT
Ideal
numba_pt
openmp_best
Tiempo (s)
Threads
100
80
60
40
20
1
2
4
8

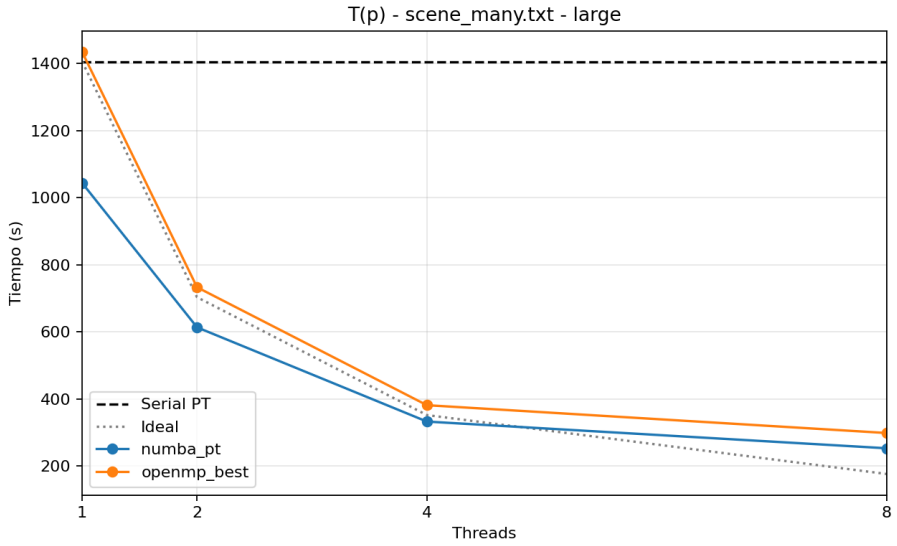

Speedup - scene.txt - medium
numba_pt speedup
openmp_best speedup
Ideal
Speedup
Threads
1
2
3
4
5
6
7
8
1
2
4
8

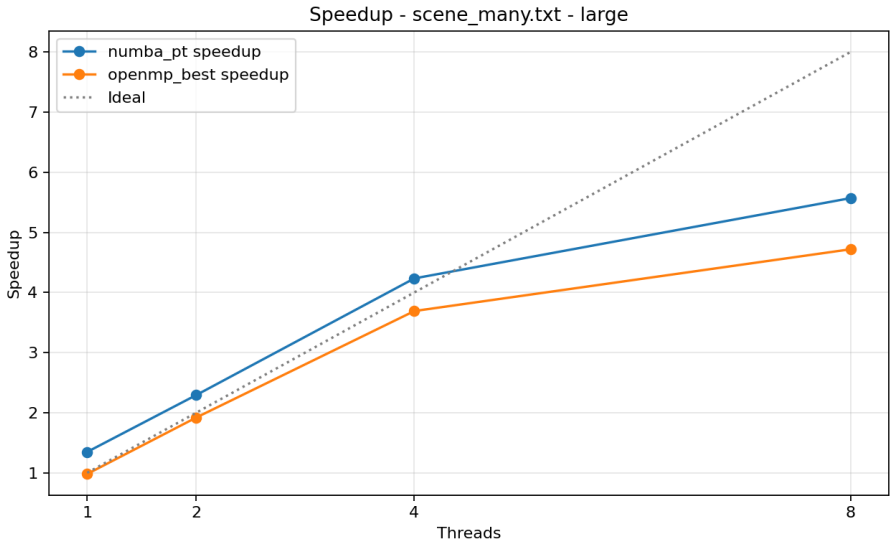

Eficiencia - scene.txt - medium
numba_pt
openmp_best
Ideal
Eficiencia
Threads
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1
2
4
8

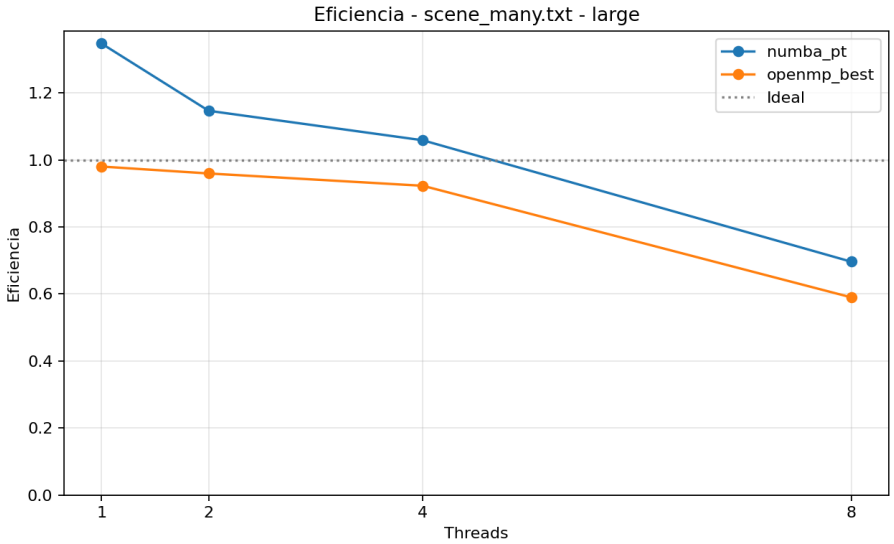

T(p) - scene.txt - small
Tiempo (s)
Threads
Serial PT
Ideal
numba_pt
openmp_best
1
2
3
4
5
6
1
2
4
8

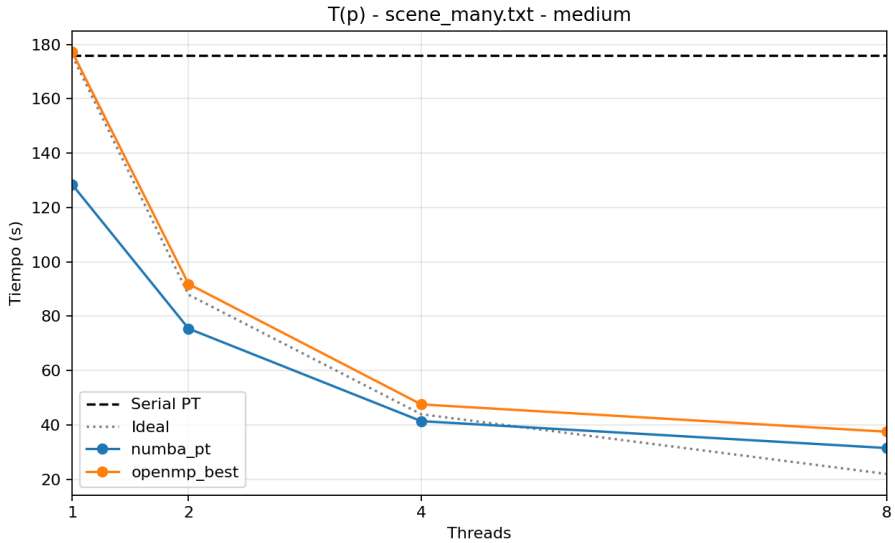

Speedup - scene.txt - small
Speedup
Threads
numba_pt speedup
openmp_best speedup
Ideal
1
2
3
4
5
6
7
8
1
2
4
8

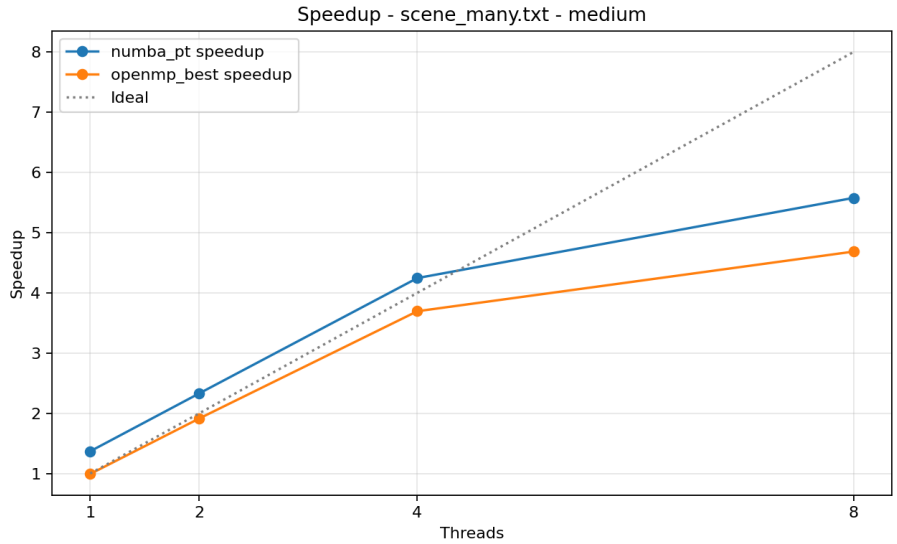

Eficiencia - scene.txt - small
numba_pt
openmp_best
Ideal
Eficiencia
Threads
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1
2
4
8

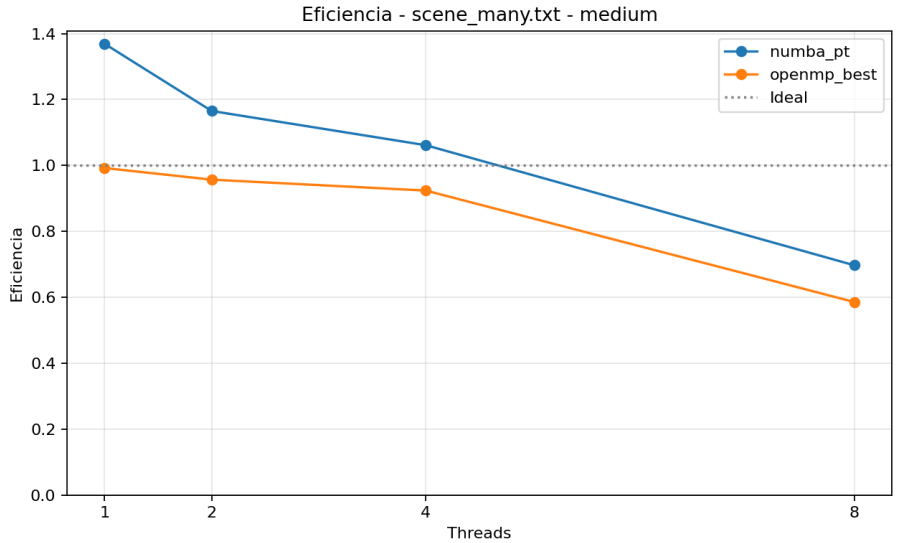

T(p) - scene_many.txt - large
Serial PT
Ideal
numba_pt
openmp_best
Tiempo (s)
Threads
200
400
600
800
1000
1200
1400
1
2
4
8

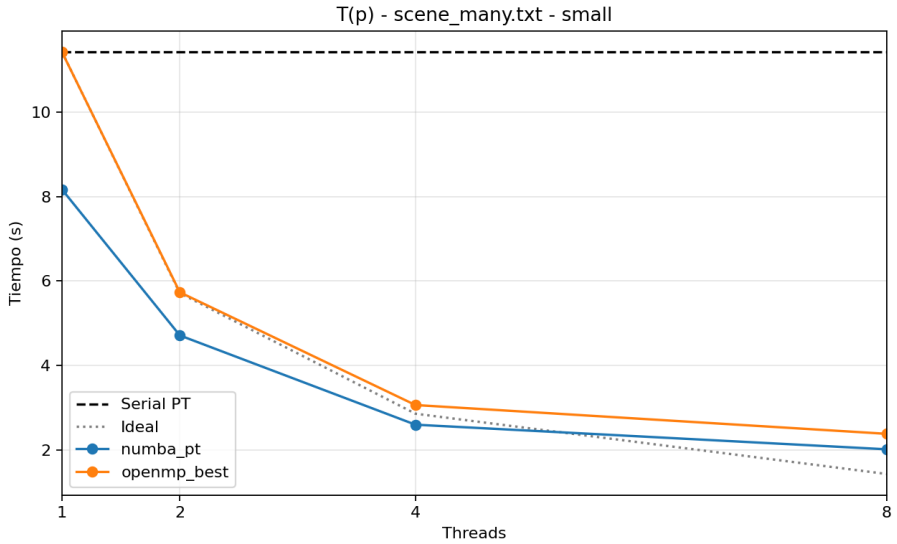

Speedup - scene_many.txt - large
numba_pt speedup
openmp_best speedup
Ideal
Speedup
Threads
1
2
3
4
5
6
7
8
1
2
4
8

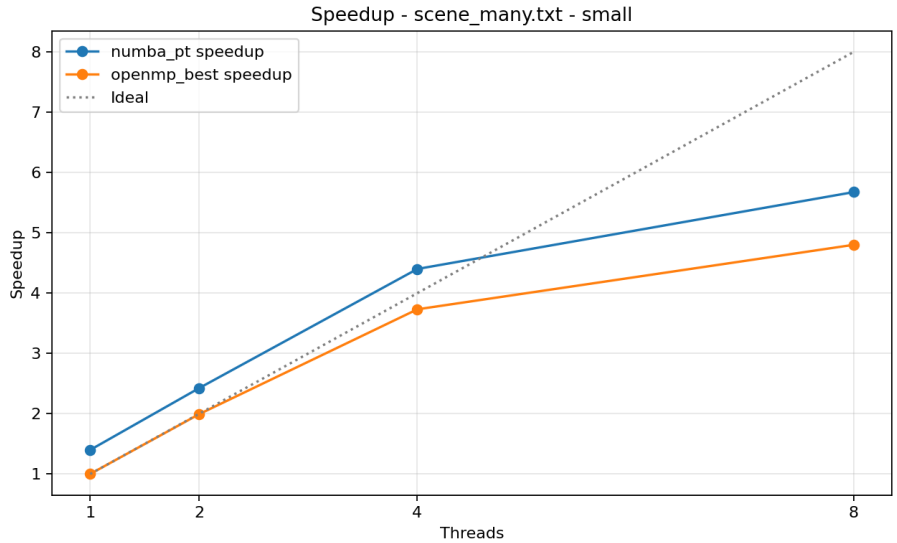

Eficiencia - scene_many.txt - large
numba_pt
openmp_best
Ideal
Eficiencia
Threads
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1
2
4
8

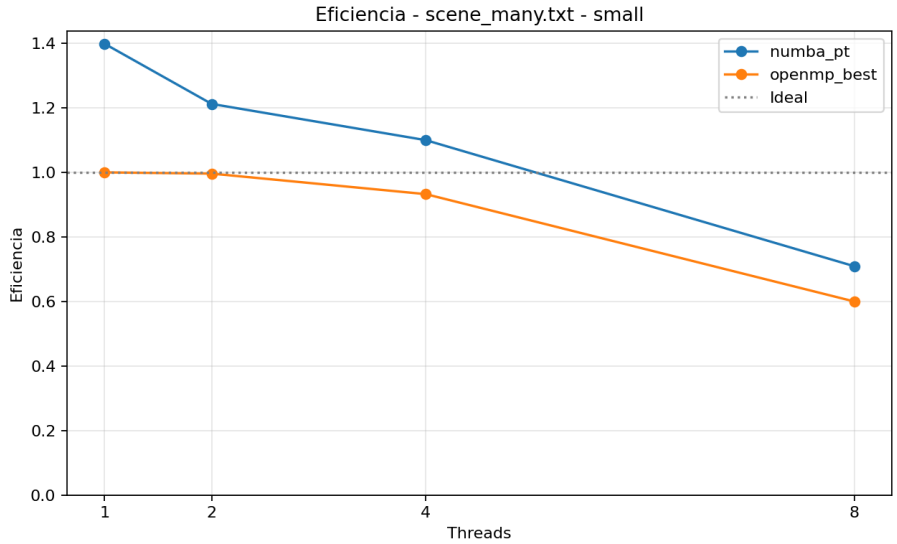

T(p) - scene_many.txt - medium
Tiempo (s)
Threads
Serial PT
Ideal
numba_pt
openmp_best
180
160
140
120
100
80
60
40
20
1
2
4
8


Speedup - scene_many.txt - medium
Speedup
Threads
numba_pt speedup
openmp_best speedup
Ideal
8
7
6
5
4
3
2
1
1
2
4
8


Eficiencia - scene_many.txt - medium
numba_pt
openmp_best
Ideal
Eficiencia
Threads
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1
2
4
8


T(p) - scene_many.txt - small
Serial PT
Ideal
numba_pt
openmp_best
Tiempo (s)
Threads
10
8
6
4
2
1
2
4
8





Comentarios

El crecimiento esperado al duplicar la resolucion es cercano a 4x, porque se cuadruplica el numero de pixeles. En path tracing, duplicar N duplica aproximadamente el trabajo por pixel, salvo variaciones por terminacion anticipada y efectos de cache/runtime. La escena con 40 esferas aumenta K y por tanto el costo de cada interseccion, lo que vuelve mas visible el desbalance de carga.

La eficiencia decae al aumentar p por overhead fork-join, barreras implicitas,

planificacion dinamica, contencion de recursos compartidos y saturacion de partes del procesador. En esta tarea no hay reducciones ni escrituras compartidas sobre una misma variable; el principal riesgo de rendimiento es el desbalance y el costo de scheduling con chunks demasiado finos.

Segundo computador

Ejecute el mismo comando completo en otro computador y agregue aqui la tabla y graficos generados. Deben declararse cores fisicos, threads logicos y frecuencia del procesador de ese equipo.